



CITY COLLEGES

BUSINESS INTELLIGENCE & DATA WAREHOUSING CA2

“Designing a Retail Analytics Solution Using BigQuery”

“Mid-Sized Retail SuperStore”

Karina De La Luz Sanchez

13th Jan, 2026



Contents

Background.....	4
Required Sections.....	4
1. Business Overview & Analytical Objectives	7
1.1 Business Overview	7
1.2 Operational Considerations	7
1.3 Analytical Objectives and Business Questions	8
2. Data Sources & Ingestion Strategy	10
2.1 Data Sources	10
2.2 Data Structure and Format	10
2.3 Data Ingestion Approach	11
2.4 Update Frequency and Data Refresh	11
2.5 Rationale for Using BigQuery Sandbox	12
3. Data Modeling	13
3.1 Conceptual Data Model	13
4. ETL / ELT Pipeline Design	17
4.1 ETL / ELT Approach	17
4.2 Source to Staging Layer	17
4.3 Data Transformation Logic	18
4.4 Fact Table Loading	20
4.5 Sequence of Loading Activities.....	21
4.6 Data Quality and Validation Considerations	21
5. Cloud Architecture Using Google Cloud Platform (GCP)	22
5.1 Cloud Architecture Overview	22
5.2 Architecture Components and Their Roles	22
5.3 Access Control and Security.....	24
5.4 Compute, Storage, and Cost Considerations	24
5.5 Architecture Benefits	24
6. Looker Studio Dashboard.....	25
6.1 Dashboard Objectives	25



6.2 Data Connection and Model Applications	25
6.3 Key Visualisations and Metrics	25
6.4 Interactivity and Filters.....	27
6.5 Dashboard Layout and Design Considerations	28
6.6 Screenshots and Documentation.....	28
6.7 Dashboard Insights & Business Interpretation	28
7. Data Governance, Metadata & Data Quality	30
7.1 Data Governance Considerations.....	30
7.2 Metadata Management and Data Lineage.....	30
7.3 Data Quality Rules and Validation	30
7.4 Data Privacy and Security	31
7.5 Master Data Management Approach	31
8. Future Predictive Analytics Opportunities.....	32
8.1 Sales Forecasting	32
8.2 Customer Churn and Segmentation Prediction	32
8.3 Infrastructure and Data Readiness	33
8.4 Strategic Impact.....	33
Conclusion	34
Appendices	35
Appendix A: Complete Looker Studio Dashboard	35
Appendix B – SQL Scripts for Data Warehouse Creation	36
References	39



Background

You are acting as an analytics consultant for a mid-sized retail company. The company wants to modernise its data analytics capability using **Google BigQuery** and develop data-driven insights to support business decision-making.

Your task is to design an **end-to-end analytics solution** that includes data ingestion, modelling, warehousing, cloud architecture, BI reporting, governance, and future analytics opportunities.

Required Sections

Your assignment is to create a comprehensive data management strategy document that addresses the following components

1. Business Overview & Analytical Objectives (10%)

Provide:

- A profile of the retail organisation you have chosen
- A description of the key operational challenges it faces
- A set of analytical questions the company wants to answer using data

2. Data Sources & Ingestion Strategy (10%)

Identify:

- All internal and external data sources relevant to retail operations
- The structure and format of each source
- The expected update frequency
- Your approach for ingesting these datasets into BigQuery

3. Data Modeling (15%)

Develop:

- A conceptual model for the organisation
- A logical model showing relevant entities and relationships
- A star schema suitable for analytics, including at least one fact table and three dimension tables

You must clearly describe the design choices you make for your schema.



4. ETL/ELT Pipeline Design (10%)

Explain:

- How data will be moved from source systems into a staging layer
- How it will be transformed for analysis
- How it will be loaded into the final warehouse tables
- The sequence of loading activities

Include indicative SQL for key transformation or loading steps.

5. Cloud Architecture Using GCP (10%)

Present:

- A high-level architecture showing how your solution will be implemented using Google Cloud components
 - A description of the role of each component
 - An explanation of how storage, compute, access control, and cost considerations are managed
- A diagram is required.

6. Looker Studio Dashboard (15%)

Create:

- A functional dashboard connected to BigQuery
- A set of visualisations that answer the analytical questions you defined earlier
- A clear, well-structured layout that supports decision-making
- At least one interactive filter

You must include screenshots in your report.

7. Data Governance, Metadata & Data Quality (10%)

Outline:

- Governance considerations relevant to retail data
- How metadata and lineage will be managed
- Key data quality rules
- How sensitive or personal data will be protected
- Your approach to managing master data across the organisation



8. Future Predictive Analytics Opportunities (10%)

Identify:

- One or more opportunities to apply predictive analytics in the future
- The business value these initiatives would provide
- The additional data, features, or infrastructure they would require

9. Presentation, Structure & Referencing (5%)

Your submission will also be assessed on:

1. Writing quality and clarity
2. Structure and coherence
3. Use of diagrams and tables
4. Correct referencing style



1. Business Overview & Analytical Objectives

1.1 Business Overview

The chosen company for the study is a mid-sized retail superstore with multiple locations. The group provides scaling ranges of products in three main categories- Furniture, Office Supplies and Technology. These classes are indicative of the product offerings mix for an average superstore in which both retail consumers and small-medium business customers are served.

The company primarily sells its product at retail through its chain of stores, with a centralized purchasing and distribution system. For every sale, the order information is captured; this includes product specifications, sales' amount values, discount ratios as well as shipping and profitability information. This data has expanded over time, and today it's hard to gather stakeholder insights in a timely manner with traditional reporting tools.

Increasing competition in retail has highlighted the need for the company to update its analytics capabilities. Management wants a cloud-based data warehouse and BI platform enabling easy storage scaling, analytics and data driven decision making. As an analytics consultant, the scope of this project is to build a full-stack retail analytics solution on Google BigQuery and Looker studio to meet these objectives.

1.2 Operational Considerations

The organisation is experiencing several operational and analytical challenges despite the availability of vast amounts of transactional data.

First, profitability differs a lot by product, region and customer type. Even if some products sell well, profit margins might be minimal or even negative due to discounting too much or paying a premium for shipping. With no unified analytics solution, decision makers can hardly see the forest for the trees - which are actually their products and regions that contribute to a sustainable profitability.

Secondly, the company finds it difficult to comprehend trend information about sales performance over time. Now previous transactional level sales data and offer inventory levels are available for you to uncover patterns of seasonality, growth or declining products or categories. This limitation prevents organisation from planning promotions, managing inventory well and forecasting future demands.

Lastly, reporting is overwhelmingly manual and siloe Data is housed in flat files and NetRep spreadsheets that require tedious preparation before analysis can be conducted. This causes lags in reporting and higher risk of lack consistency in management reports.

These difficulties in the system demonstrate the necessity for a centralised data warehouse and a carefully crafted analytical model that enables instant, uncharged, and reliable insights.



1.3 Analytical Objectives and Business Questions

The main analysis feature of this project is to help the retail company to gain actionable insights from their transactional data via a cutting-edge analytics platform.

In particular, the solution should be able to respond the following main analytical questions:

- What are the top sales and profit products categories and subcategories?
- How do sales volume and profit respond with discounts?
- What are the revenue and profit shared between various types of customers?
- What are the monthly and yearly sales profit trends?

It is with these questions that management can obtain a deeper understanding of the performance of its business, pinpoint inefficiencies and drive important strategic pricing, promotion, inventory and regional expansion decisions.

These insights will be surfaced via a modelled data warehouse in BigQuery and an interactive dashboard in Looker Studio, giving stakeholders a powerful and intuitive view of retail performance.

The main aim of this analytics solution is to help the retailing company in converting raw transactional data into useful and dependable business insights that can be utilized strategically, tactically and operationally.

More specifically, the DW and BI platform will be built taking into account the following analytics needs:

- Enhanced sales and profitability drivers visibility
- Find non-performing products, areas and fragments of customers.
- Optimise pricing and discount strategies
- Add to your inventory planning and demand forecasting efforts
- Facilitate the consistent and automated production of management reports

The aforementioned high level goals can be achieved by addressing the below-mentioned core abatement business questions:

Sales & Profitability Performance

- What are the top and bottom producing product categories and sub-categories?
- Which specific products most frequently contribute to profit, and which cause the losses?
- What are the regional, state and city differences in profitability?



Pricing & Discount Strategy

- How do discounts relate to profit margins?
- At what discount levels does profitability erode materially?
- What types of products or product categories are discount-sensitive?

Customer Analytics

- What are the share of revenue and profit from each customer segment (Consumer, Corporate, Home Office)?
- What segments are the most profitable among customers with the highest AOV and LTV?

Time-Based Trends

- How do sales and profit evolve monthly and yearly?
- Are there identifiable seasonal patterns or recurring peak periods?
- Which products show consistent growth or decline over time?

Answers to these questions will enable management to:

- Optimise product portfolio decisions
- Enhance regional growth and logistic scheduling
- Design more effective promotional strategies
- Optimize marketing and operational spend more effectively

These insights come to life via a dimensional data warehouse in Google BigQuery, interactive Looker Studio dashboard and near real-time access to accurate, trusted and business-read analytics.

2. Data Sources & Ingestion Strategy

2.1 Data Sources

The historical transactional internal data from a midsize retail superstore is considered as the main source in this analytics tool. Detailed information on customer orders is accumulated over years which facilitates the analysis of sales performance, margin and the operator's efficiency.

The data set contains also transaction, product, customer, geo and time attributes. Every record is a single sales transaction at the product level so it has great granularity for analytical processing and reporting.

Table 1 below presents an overview of the main data sources used in the retail analytics solution.

Table 1. Retail Data Sources

Data Source	Description	Format	Update Frequency
Sales Transactions	Order-level sales data including revenue, profit, discounts, and shipping cost	Excel / CSV	Daily (batch)
Product Data	Product category, sub-category, and product name	Excel / CSV	Weekly
Customer Data	Customer identifiers, names, and segments	Excel / CSV	Daily
Location Data	Geographic attributes including country, region, state, city, and postal code	Excel / CSV	Weekly
Date Data	Order and shipping dates used for time-based analysis	Derived	Daily

Even though the data set is readily given in an Excel file for analytical use, it is technically treated as several logical data sources to model how they would normally be produced across operational systems within any real retail company.

2.2 Data Structure and Format

The dataset is in a tabular form, where each row represents an individual sales transaction. The data consists of numeric, categorical, and date attributes.



Key structural characteristics include:

- **Numerical fields** such as sales amount, profit, quantity sold, discount, and shipping cost.
- **Categorical fields** including product category, customer segment, and geographic region.
- **Date fields** such as order date and ship date, which support time-based analysis.

Before upload to the data warehouse in being converted from Excel to CSV to be compatible with Google BigQuery's batch upload feature.

2.3 Data Ingestion Approach

The analytics portion uses an ELT (Extract, Load, Transform) process that matches incredibly well with cloud-first data warehouse solutions as in the case of BigQuery.

The process of ingestion is as follows:

1. **Extract**

The transactional data set pulls from the source system into Excel and is exported as a CSV file.

2. **Load**

The raw CSV is uploaded to Google BigQuery under the sandbox environment and staged into a table. This is a table in which the data are still kept raw to carry out lineage and traceability.

3. **Transform**

Data prep happens right in BigQuery with SQL. These operations are type conversions, derivation of computed fields and load the data into dimension and fact table to analyze the data.

This ELT approach reduces the amount of data movement, utilises BigQuery's compute driven design and simplifies the pipeline architecture.

2.4 Update Frequency and Data Refresh

In a live system, you'd probably want to ingest transactional sales data as a daily batch layer in order not to affect your reporting cadence while still providing efficient operation. Dimension data – ie details about products, locations etc – would be updated less frequently (eg weekly) as the data in these dimensions evolves more slowly.



As this project, the data set is consumed as a point in history that is loaded once into BigQuery Sandbox. However, ingestion approach has the capability to extend into incremental data load effortlessly in future.

2.5 Rationale for Using BigQuery Sandbox

Google BigQuery Sandbox was chosen as the data warehousing platform for its serverless architecture, scalability, and native integration with Looker Studio. The sandbox environment also provides a blueprint for creating a real-world cloud-based analytics solution that can be used for education and prototyping without needing to worry about provisioning infrastructure or managing costs.

This approach allows the Organization to homogeny their retail data, do transformations in a secure fashion across all that information tent and then deliver meaningful insights through an interactive dashboard.



3. Data Modeling

3.1 Conceptual Data Model

At a conceptual level, the data model of our retail organization is formulated to capture the core business mid-processes and entities taking part in retailing. The basic business happening that the system documents is a sales transaction, at which point a customer buys any number of products from the megastore.

In the conceptual schema, the main entities are listed as below:

- **Sales:** Represents the transactional event, capturing financial and operational metrics.
- **Customer:** Represents individuals or organisations purchasing products.
- **Product:** Represents items sold by the retailer, organised into categories and sub-categories.
- **Location:** Represents the geographic context in which sales occur.
- **Date:** Represents the temporal aspect of sales transactions.

The conceptual model focuses on the necessary data for operation, not how it is stored. It offers an overview of organization operations at a high level and gives a background for further logical or physical data models.

3.2 Logical Data Model

The logical data model translates the conceptual view into a structured representation of entities, attributes, and relationships. At this level, the focus is on defining how entities relate to one another while remaining independent of any specific database technology.

In the logical model:

- Each **sales transaction** is between precisely one customer (or no customer), one product, one location and one order date.
- A **customer** can produce many different orders over time.
- A **product** may take part in a lot of different sales transactions.
- A **location** is associated with many sales transactions--but the most are clustered in different regions or cities.
- A **date** can belong to many transactions, allowing time-based aggregation of results.

For each entity, unique primary keys are defined to identify records, and foreign keys are used to represent relationships between entities. This logical structure ensures data consistency and allows flexible analytic queries.

This structure guarantees referential integrity and supports efficient analytic joins.

3.3 Dimensional Modeling Approach

A dimensional modeling approach is adopted to support analytical reporting and business intelligence requirements. Specifically, a star schema is chosen for its ease of use, high performance efficiency, and ability to aggregate large quantities of transaction data.

The star schema comprises:

- One central **fact table** storing numerical measures.
- Multiple **dimension tables** providing descriptive context for analysis.

This makes querying simple, dashboard development intuitive inside Looker Studio, and is best-practice data warehousing for retail analytics.

3.4 Star Schema Design

The proposed star schema for the retail analytics solution is illustrated below.

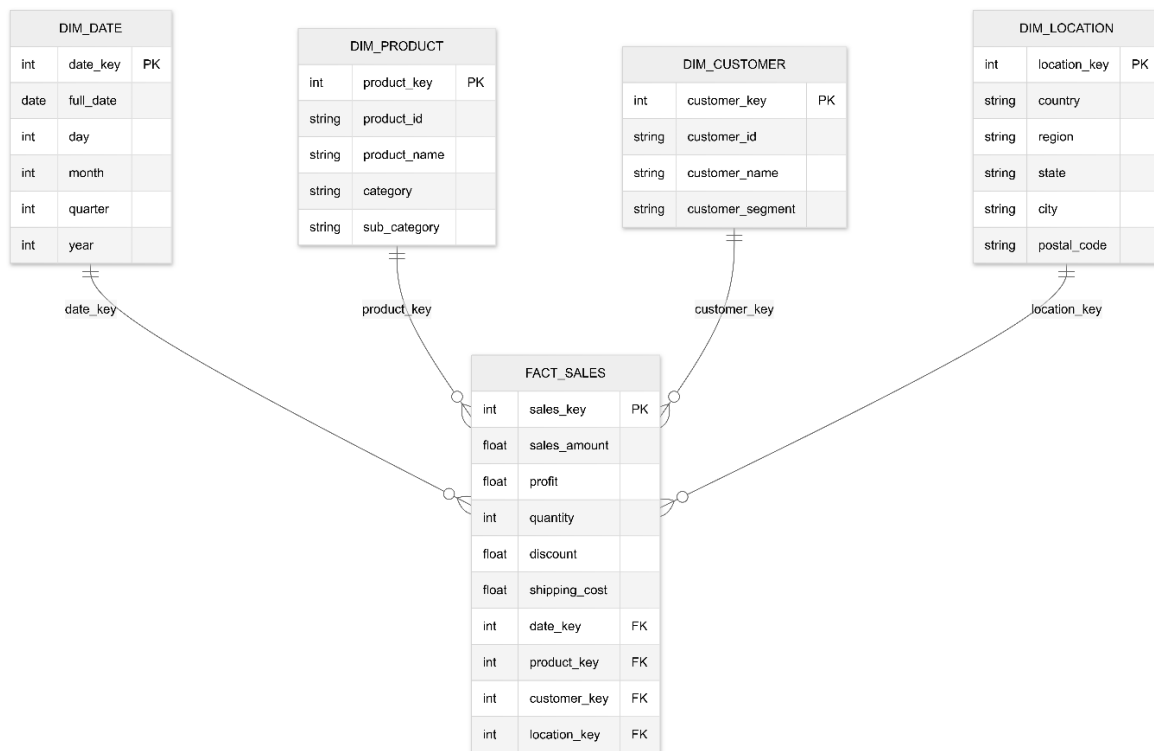


Figure 1. Star Schema Data Model for Retail Sales Analytics¹

¹ Diagram generated using Mermaid.js, a text-based diagramming tool (<https://www.mermaidchart.com>).

Fact Table

Fact_Sales

The Fact_Sales table captures measurable business metrics at the transaction level. Each row represents one product sold within an order.

Key measures include:

- Sales amount
- Profit
- Quantity ordered
- Discount applied
- Shipping cost

Foreign keys link the Fact_Sales table to the relevant dimensions, thus allowing multi-dimensional analysis.

Dimension Tables

Dim_Date

Stores time-related attributes derived from the order date, including:

- Date key
- Full date
- Day
- Month
- Quarter
- Year

This dimension helps with trend analysis, seasonality detection and time-based aggregations.

Dim_Product

Contains descriptive product attributes, including:

- Product key
- Product name
- Product category
- Product sub-category

This dimension permits sales and profitability analysis across product hierarchies.



Dim_Customer

Stores customer-related information such as:

- Customer key
- Customer ID
- Customer name
- Customer segment

This dimension allows customer segmentation and supports revenue contribution analysis.

Dim_Location

Captures geographic attributes, including:

- Location key
- Country
- Region
- State or province
- City
- Postal code

This dimension allows performance comparison across regions and supports geographic drill-down in dashboards.

3.5 Design Justification

The star schema design was chosen to balance analytical flexibility with performance efficiency. By separating quantitative measures from descriptive attributes, the model supports fast aggregation and simplifies query logic.

The granularity of the Fact_Sales table is defined at the transactional and product level so that a detailed analysis can be carried out and higher-level summaries obtained through aggregation. This level of granularity is particularly important in retail analytics. Profits and discounts often vary significantly by product.

Overall, this data model offers a rock-solid basis for business intelligence reporting, is easy to scale for the future, and conforms with industry-standard data warehousing practices.



4. ETL / ELT Pipeline Design

4.1 ETL / ELT Approach

The retail analytics solution adopts a process as stated ELT (Extract, Load, Transform) pipeline architecture, which is good practice for cloud data warehouses like Google BigQuery.

This approach first loads raw data into the data warehouse environment and then transforms it using SQL there.

With this design, BigQuery's scalable compute engine takes full advantage of. It simplifies pipeline maintenance by reducing unnecessary data movement. It is particularly applicable to analytical workloads involving large quantities of historical transactional data, such as retail sales.

4.2 Source to Staging Layer

Data Extraction

The source data comes from our internal retail transaction system, provided in Excel format. Before it can be ingested into BigQuery, however, the file must be converted into CSV format--this will make it compatible to batch loading processes. You just have to imagine that part.

Staging Layer Design

The CSV file is loaded as a staging table into BigQuery. The table holds this information in raw form and serves as an intermediate layer connecting the source system and the analytical warehouse.

The staging layer has many functions:

- Preserves original data for traceability and auditing
- Supports data validation and quality checks
- Enables repeatable transformations without reloading source files

The screenshot shows the BigQuery interface for the 'superstore_raw' table. The left sidebar displays a project tree with 'superstore_raw' selected. The main panel shows the 'Schema' tab for the table. A filter bar is present above the schema table. The schema table lists the following fields:

Field name	Type	Mode	Description	Key	Collation	Default value	Policy
Row ID	INTEGER	NULLABLE	-	-	-	-	-
Order Priority	STRING	NULLABLE	-	-	-	-	-
Discount	FLOAT	NULLABLE	-	-	-	-	-
Unit Price	FLOAT	NULLABLE	-	-	-	-	-
Shipping Cost	FLOAT	NULLABLE	-	-	-	-	-
Customer ID	INTEGER	NULLABLE	-	-	-	-	-
Customer Name	STRING	NULLABLE	-	-	-	-	-
Ship Mode	STRING	NULLABLE	-	-	-	-	-
Customer Segment	STRING	NULLABLE	-	-	-	-	-

Buttons at the bottom: [Edit schema](#) and [View row access policies](#).

Figure 2. Raw transactional data loaded into the staging table (superstore_raw) in BigQuery.

4.3 Data Transformation Logic

Data transformations take place directly in BigQuery using SQL. The transformation logic concentrates on:

- Standardising data types
- Removing duplicate records
- Using deterministic row numbering logic
- Preparing dimension and fact tables aligned with the star schema

Dimension Table Transformations

Each dimension table is populated by extracting distinct values from the staging layer. (see Figure 3 and 4 below).

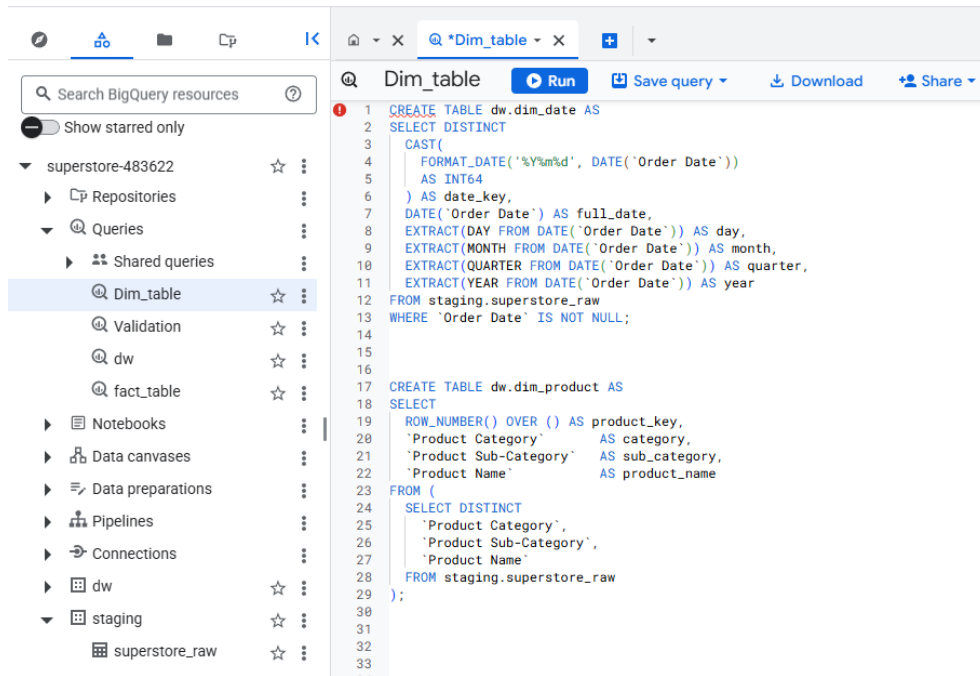


Figure 3. Creation of Date and Product dimension tables in BigQuery.

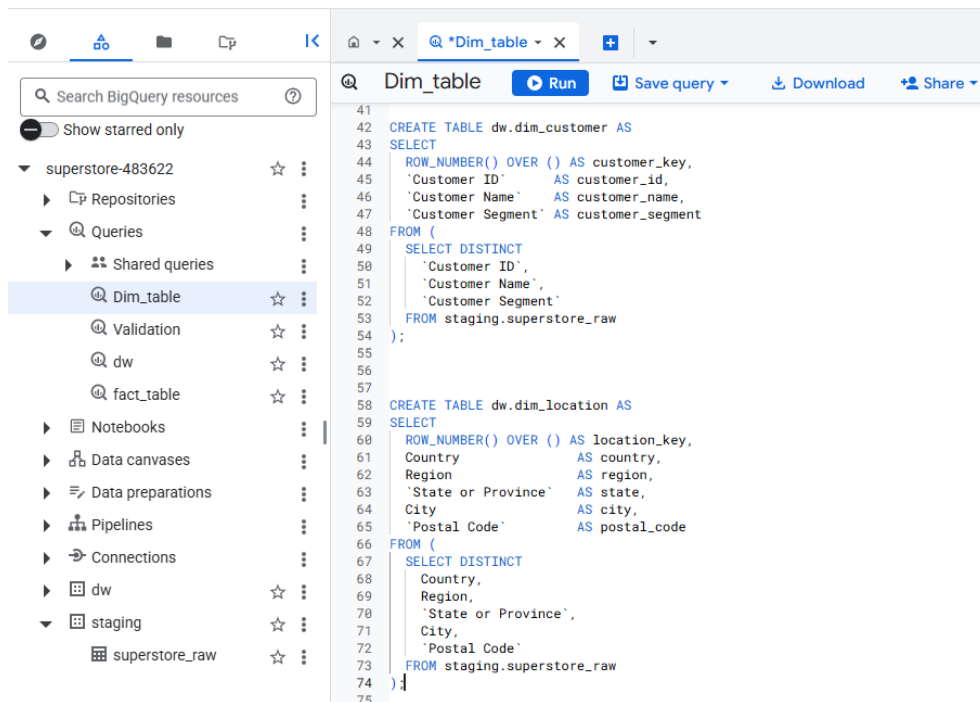
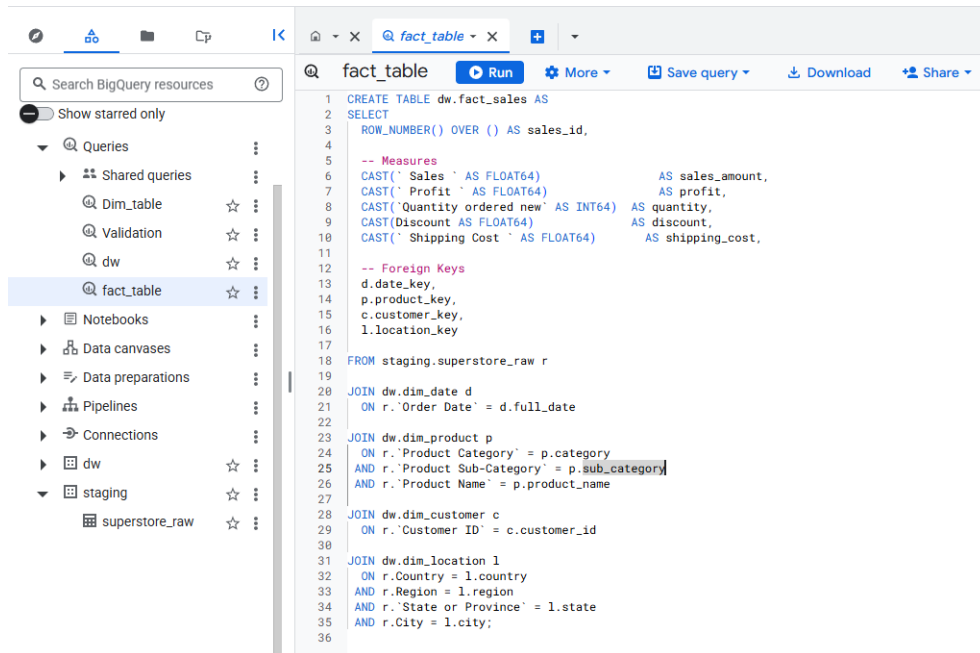


Figure 4. Creation of Customer Dimension and Location Dimension tables on Bigquery.

4.4 Fact Table Loading

After all dimension tables have been constructed, the fact table is filled. This ensures that foreign key relationships can be properly resolved.

The grain of the fact table defines one sale per product sold within an order, which defines the grain of the fact table.



The screenshot displays the Google Cloud BigQuery console interface. On the left, a sidebar shows a project tree with folders for 'Queries', 'Notebooks', 'Data canvases', 'Data preparations', 'Pipelines', 'Connections', and a 'dw' dataset. The 'fact_table' query is selected and highlighted. The main panel shows the SQL query for creating the 'fact_sales' table. The query includes a 'CREATE TABLE' statement followed by a 'SELECT' statement that joins the 'staging.superstore_raw' table with several dimension tables ('dw.dim_date', 'dw.dim_product', 'dw.dim_customer', 'dw.dim_location') using their respective surrogate keys. The 'SELECT' statement also includes measures like 'Sales', 'Profit', 'Quantity ordered new', 'Discount', and 'Shipping Cost', all cast to appropriate data types.

```
1 CREATE TABLE dw.fact_sales AS
2 SELECT
3   ROW_NUMBER() OVER () AS sales_id,
4
5   -- Measures
6   CAST(' Sales ' AS FLOAT64) AS sales_amount,
7   CAST(' Profit ' AS FLOAT64) AS profit,
8   CAST(' Quantity ordered new' AS INT64) AS quantity,
9   CAST(' Discount ' AS FLOAT64) AS discount,
10  CAST(' Shipping Cost ' AS FLOAT64) AS shipping_cost,
11
12  -- Foreign Keys
13  d.date_key,
14  p.product_key,
15  c.customer_key,
16  l.location_key
17 FROM staging.superstore_raw r
18
19 JOIN dw.dim_date d
20   ON r.'Order Date' = d.full_date
21
22 JOIN dw.dim_product p
23   ON r.'Product Category' = p.category
24  AND r.'Product Sub-Category' = p.sub_category
25  AND r.'Product Name' = p.product_name
26
27 JOIN dw.dim_customer c
28   ON r.'Customer ID' = c.customer_id
29
30 JOIN dw.dim_location l
31   ON r.Country = l.country
32  AND r.Region = l.region
33  AND r.'State or Province' = l.state
34  AND r.City = l.city;
```

Figure 5. Creation of the Sales fact table (fact_sales) in BigQuery.

The full set of SQL scripts used to create and populate the dimension and fact tables is provided in **Appendix B – SQL Scripts for Data Warehouse Creation**. These scripts illustrate the practical implementation of the ELT process in BigQuery, including surrogate key generation, data cleansing, and referential integrity management.



4.5 Sequence of Loading Activities

The ETL / ELT pipeline has a structured sequence in order to preserve data integrity:

1. Load raw data into the staging table
2. Fills dimension tables (Date, Product, Customer, Location)
3. Check dimension data and key generation
4. Load the fact table using foreign key lookups
5. Perform basic validation checks (row counts, null keys)

This order of stages maintains referential integrity and ensures that consistent analytical queries can be performed.

4.6 Data Quality and Validation Considerations

During the transformation process basic data quality checks are carried out, including:

- Ensuring no null values exist in surrogate keys
- Validating that sales and quantity values are non-negative
- Verifying referential integrity between fact and dimension tables

These checks improve the accuracy of the analytical outputs and reduce the risk of misleading findings.

5. Cloud Architecture Using Google Cloud Platform (GCP)

5.1 Cloud Architecture Overview

The provided retail analytics solution is implemented using a cloud-native architecture based on Google Cloud Platform (GCP). It leverages fully managed products to support high scalability, strong performance and low maintenance complexity while removing operational problems altogether.

This architecture adopts an approach of layering, taking data from input and storage all the way through to analytics visualization. This not only provides a clear separation of concerns, but also conforms with best practices for modern data warehouses and business intelligence.

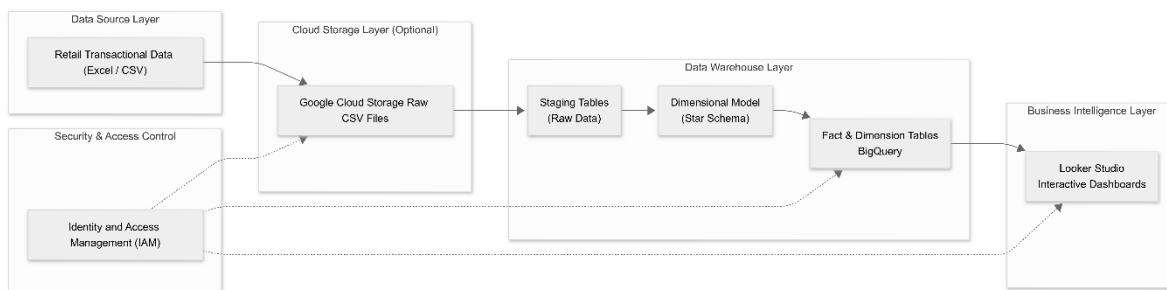


Figure 6. High-level Cloud Architecture for the Retail Analytics Solution on Google Cloud Platform²

The architecture represents a cloud-native analytics pipeline on Google Cloud Platform where historical retail data is ingested, transformed using BigQuery and visualised through Looker Studio. It is protected by Identity & Access Management (IAM).

5.2 Architecture Components and Their Roles

Data Source Layer

The solution begins with internal retail transactional data provided in Excel format from the Kaggle dataset repository. In a real-world environment, this data would originate from operational systems such as point-of-sale (POS) platforms or order management systems.

In a real environment, this data would come from operational systems such as POS systems or order management systems.

² Diagram generated using Mermaid.js, a text-based diagramming tool (<https://www.mermaidchart.com>).

Figure 7. Superstore retail transactional dataset (source data).

- 
- Real-time querying of warehouse data
 - Interactive filters and drill-down analysis
 - Visualisation of key performance indicators (KPIs)

By separating data storage from data visualisation this model provides business users ability to query without disturbing lowerorder data models.

5.3 Access Control and Security

Access control within the architecture is managed using identity and access management (IAM) principles. Permissions are given on the basis of user roles to ensure only authorized users can obtain or modify data. Key security points include:

- Restricting write access to staging and warehouse tables
- Granting read-only access for reporting users
- Protecting sensitive customer information through controlled access and data minimisation.

This method complies with data governance requirements and permits safe sharing of data across the enterprise environment.

5.4 Compute, Storage, and Cost Considerations

By separating the storage from its processor, BigQuery has helped the organisation save money by only having to pay for what it consumes. Annual storage costs are relatively low and CPUs cost by the hour, which allows dynamic inflation according to query complexity and frequency.

Using the BigQuery Sandbox for this project gives an example of a cost-effective analytics solution which can be deployed without having to foot an up-front infrastructure bill. In a production situation, cost controls could be implemented such as query quotas and dataset-level permissions.

5.5 Architecture Benefits

The proposed GCP architecture being proposed has many advantages:

- **Scalability:** Easily processes vast amounts of historical retail data
- **Performance:** Optimized for analysis tasks, seeing lots of data together
- **Simplicity:** Fully managed services reduce operational overhead
- **Flexibility:** Supports future extensions such as predictive analytics and machine learning.

Hence this architecture provides a solid platform for retail analytics solutions that is already in place and ready for the future.



6. Looker Studio Dashboard

6.1 Dashboard Objectives

The chief aim of the Looker Studio dashboard is to provide executives with a clear, interactive and consolidated view of retail performance. The dashboard has been designed to convert the analytical insights stored in the BigQuery data warehouse into easy representations on screen that can support strategic and operational decision making.

The dashboard directly responds to the analytical questions that were set out in the project brief. Coverage is focused on sales performance, profitability, customer segmentation, and geographical trends.

6.2 Data Connection and Model Applications

The dashboard is connected directly to Google BigQuery via the dimensional model built in the data warehouse. The central Fact_Sales table is joined with Date, Product, Customer and Location dimensions to allow flexible slicing and aggregation of measures.

Using BigQuery as the data source we get:

- Consistent metrics across all visualisations
- Real-time querying of warehouse data
- Separation between data processing and visualisation layers

This design supports scalability and ensures that dashboard logic remains simple and maintainable.

6.3 Key Visualisations and Metrics

The dashboard is made up of a balanced range of visualisations, and mixed low-level components and high level performance indicators.

Key Performance Indicators (KPIs)

As indicated in Figure 8, the KPI scorecards appear at the top of the dashboard give a short overview of company performance, covering:

- Total Sales
- Total Profit
- Total Quantity Sold
- Average Discount

With this information, users are able to get a quick idea of the overall performance of their organisation before looking at more detailed views.



Figure 8. Key Performance Indicators (Total Sales, Total Profit, Total Quantity Sold, and Average Discount).

Sales and Profit Analysis

In order to support product and category-level analysis, the dashboard includes:

- A **bar chart** showing total sales by product category (Furniture, Office Supplies, Technology)
- A **column chart** displaying profit by product sub-category

These visualisations help identify which product groups drive revenue and which contribute most to overall profitability, as illustrated in Figure 9.

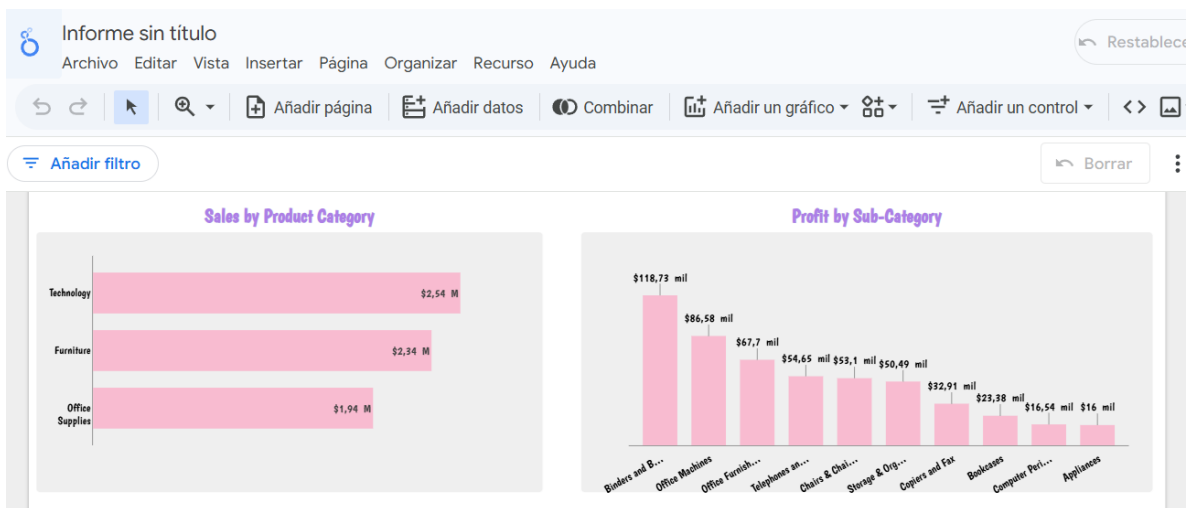


Figure 9. Total Sales by Product Category and Profit by Product Sub-Category.

Time-Based Trends

Use a time series chart like that shown in Figure 10 to illustrate monthly sales trends which are modeled over several years. This visualisation is capable of:

- Identification of seasonality patterns
- Detection of growth or decline trends
- Comparison of performance across time periods

Geographic Performance

A geographic visualisation is used to analyse regional performance:

- A **map** displaying sales or profit by region and state (see figure 10)

This view allows users to compare performance across locations and identify high- and low-performing regions.

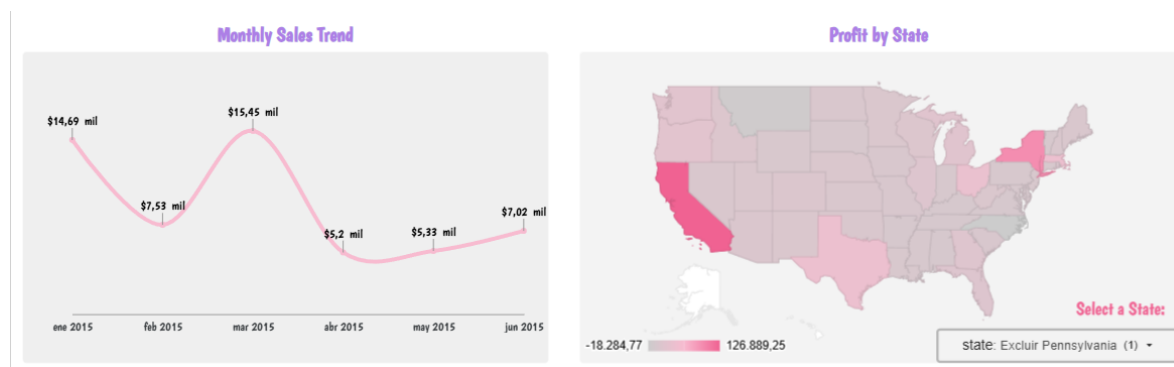


Figure 10. Monthly Sales Trend Analysis and Geographic Distribution of Profit by State

6.4 Interactivity and Filters

Interactivity is a central philosophy of dashboard design. Users might use two active filters of Looker Studio to change their analysis according their own demands (see Figure 11).

The filters consist of:

- Date range filter
- Region filter

These controls enable users to dynamically explore the data, with no new queries or reports. This is a plus for both usability and decision making ability.



Figure 11. Interactive Date and Region Filters in Looker Studio Dashboard

6.5 Dashboard Layout and Design Considerations

The dashboard follows a clear and structured layout for readability and ease of use. A proper reading flow is established to lead users from high-level KPIs down to more detailed analyses.

Planning considerations include:

- Consistent use of colours and labels
- Logical grouping of related visualisations
- Avoidance of visual clutter

The overall design balance is of executive-level insights with analytical depth, making it suitable for both senior management and operational users.

6.6 Screenshots and Documentation

Images of the completed Looker Studio dashboard have been added to this report in order to demonstrate the main analytical perspectives and interactive functions.

Moreover, all visualizations on which the complete solution was based are shown directly in **Appendix A: Complete Looker Studio Dashboard Layout & All Implemented Visuals** to allow interested readers check out in detail how it works in practice.

6.7 Dashboard Insights & Business Interpretation

This study provides dashboard insights about the Business Interpretation

Not only does the dashboard deliver a visual representation of retail performance, but it helps management gain actionable business insight.

By examining the KPI, we can see that the business made **\$6.82m in total sales** and generated a **profit of \$491,000** in total for the period; sound volume however as always there are pinpoint areas ripe for optimising –namely discounts management where an excessive **4.92%** average discount rate is being applied across deals.



At product level, the Technology category is associated with highest revenue sales followed by Furniture and Office Supplies. But if you look at the profit by sub-category chart, we can see that high sales does not necessarily mean high profits. Some sub-categories, such as Binders or Office Machines and Copiers, are more profitable but other bring less money than the sales volumes. This observation calls for a rethinking of pricing policies and promotion intensity for products with tight margins.

The monthly data series shows that there are distinct changes of the sales figures along time, with high and low peaks. This piece wise pattern is evidence for seasonal dependence of customer demand that companies can use for better inventory planning and promotional timing.

Geographic distribution and profitability The profit favourability reflected in this analysis is uneven across states. Some areas always do better than most, others make close to zero or even a loss. This conclusion verifies the requirement for local tactical strategies on logistics, marketing investment and store expansion.

In general, the dashboard allows management to go beyond descriptive reporting and begin making decisions based on data by connecting operational performance with strategic levers in pricing, product portfolio management, and regional planning.



7. Data Governance, Metadata & Data Quality

7.1 Data Governance Considerations

In retail analytics, good data governance is essential to maintain data accuracy in the system at source and maintain consistency of trust over the big retail chain.

For the proposed SuperStore analytics solution, governance is focused on ownership, access control, and standardisation of key business entities such as customers, products, locations and dates.

Clear data ownership is defined for each domain (sales, customers and products), analytics datasets in Google BigQuery are managed in a centralised way. This approach reduces misreporting risk and supports the development of scalable analytics and timescale-splitting outtakes as data volumes grow over time.

7.2 Metadata Management and Data Lineage

Metadata management is one of the key pieces in getting analytical datasets to be intelligible and reusable. In this solution, metadata has been arranged on several different levels:

- BigQuery datasets and tables include descriptive names (fact_sales, dim_product, dim_location) that clearly reflect their business meaning.
- Field names follow consistent naming conventions, improving readability and reducing ambiguity.
- The star schema design provides implicit data lineage, where fact tables are linked to dimension tables through surrogate keys.

Furthermore, Looker Studio's data source configuration gives you the visibility to understand field types, aggregations and calculated metrics that business users need when they need it most for how their KPIs are determined.

7.3 Data Quality Rules and Validation

To ensure analytical accuracy, several data quality rules are applied:

- Referential integrity is logically maintained through surrogate keys linking fact and dimension tables.
- Numeric measures such as sales, profit, and quantity are validated to prevent null, negative, or invalid values.
- Date fields are standardised using a dedicated date dimension to ensure consistent temporal analysis.

- 
- Duplicate records are avoided through controlled ingestion and transformation logic in BigQuery.

These rules help to ensure dependable reporting and reduce the danger of misleading insights.

7.4 Data Privacy and Security

Retail data sometimes contains sensitive customer information. To solve this problem, the solution uses Google Cloud's in-built security features:

- Identity and Access Management (IAM) restricts access to datasets based on user roles.
- Personally identifiable information (PII), such as customer names or identifiers, is limited to analytical necessity and not exposed beyond reporting requirements.
- Looker Studio access is controlled through Google account permissions, ensuring only authorised users can view or interact with dashboards.

This multipart approach to security helps align with common data protection practice.

7.5 Master Data Management Approach

In retail analytics, maintaining consistency across master data is essential. In this solution, therefore, master data entities (customers, products, locations and dates) are stored as separate dimension tables. This ensures:

- A single version of truth for each business entity
- Consistent filtering and aggregation across reports
- Easier maintenance and future extensibility

By separating off master data from transactional facts, the analysis platform can still fit into an existing model and is able both remain flexible and scalable according to principles of enterprise data warehousing.



8. Future Predictive Analytics Opportunities

8.1 Sales Forecasting

Sales forecasting is one of the most prospective future applications of predictive analytics for a retail company. By developing machine learning models, therefore, using historical transactional data stored in BigQuery predictive methods could be developed to forecast future sales.

This would represent a major business value by:

- Supporting inventory planning and stock replenishment decisions
- Reducing the risk of overstocking or stock shortages
- Improving cash flow management and budgeting accuracy
- Enabling proactive promotional planning during high-demand periods

For this use case to work, the company will need:

- Extended historical data covering multiple years
- Additional features such as promotional flags, holidays, marketing campaigns, and economic indicators
- Integration with Google Cloud services such as BigQuery ML or Vertex AI for model training, evaluation, and deployment

8.2 Customer Churn and Segmentation Prediction

The second opportunity is in predicting customer churn and behavioral segmentation. Predictive models could determine customers likely to cut back their commitment or discontinue purchases by analyzing frequency of purchase, order value, product preferences, time interval between transactions.

This initiative's business value includes:

- Targeted retention campaigns for high-risk customers
- Personalised marketing strategies based on predicted customer behaviour
- Increased customer lifetime value
- More efficient allocation of marketing resources

For this use case to work, the company will need:

- Enriched customer-level data, including interaction history and loyalty indicators
- Feature engineering to capture behavioural patterns
- Data pipelines to regularly refresh prediction outputs
- Visual integration of prediction results into Looker Studio dashboards



8.3 Infrastructure and Data Readiness

The current cloud-based architecture with its Google BigQuery is a solid foundation for future predictive analytics. Built in large part on Google Cloud Service, the available star schema design guarantees clean, structured data that can be directly turned into machine learning models.

In order to fully support advanced analytical and machine learning workloads, the organisation would benefit from:

- Automated data pipelines for continuous data ingestion
- A dedicated feature store for reusable predictive variables
- Model monitoring mechanisms to detect performance degradation
- Governance processes for managing model outputs and ethical data usage

8.4 Strategic Impact

By weaving predictive analytics into the company's decision-making routines, it will go from descriptive and diagnostic analytics to prescriptive, forward-looking advice. Such an evolution could make the business more competitive and efficient while also promoting data-driven long-term planning for future strategies.



Conclusion

The project demonstrated by this is a solution of the One-stop retail source is constructed and implemented based entirely on Google BigQuery. By using Looker studio in conjunction with it, we were able to make this idea a reality. It all began with raw transactional data. After a cloud-based management system was deployed on GCP, a data warehouse that stores dimensional information based on star schema was created through ELT processes involving structured hierarchical storage.

Key performance indicators, product-level insights, geographical performance analysis and time-based trend evaluation are all available from the given analytical model outcome. These capabilities provide data-based decision-making for multiple business functions including sales management and inventory planning or regional performance control

The project also addressed another crucial topic in modern analytics environments, one that cannot be overlooked – such as metadata management analytics environment requirements broadly fall into the following categories: data governance (metadata management), data quality, security. By implementing these elements correctly, systems are built with confidence to predict the same result for any operation performed on that system no matter where or when users access it

This project also pointed out important analytics areas for future study. From descriptive or retrospective analysis, since the organization has its cloud infrastructure and a structured data foundation that is difficult to tamper with, the company is well positioned for those areas of predictive analytics which are actually predictive

This solution demonstrates how cloud-based data warehousing and business intelligence technologies can transform retail data into actionable insights with a long-term strategic value.



Appendices

Appendix A: Complete Looker Studio Dashboard

The full Looker Studio dashboard built hand-in-hand with this retail analytics solution is available in an additional appendix as well. The dashboard is based on the dimensional data model which resides in Google BigQuery, for sales performance, profit analysis, customer breakdown, geography spreading and trend analysis in time.

As presented in Section 6 of this article, the dashboard provides a complete set of interactivity characteristics including KPIs, dynamic filters and multiple views to aid managerial decision making.

The dashboard serves as the main business intelligence view for stakeholders and illustrates how the proposed cloud-based analytics architecture can be utilized in practice.

Looker Studio Dashboard (Business Intelligence Layer)

This link provides access to the interactive dashboard that was created for this project.
<https://lookerstudio.google.com/reporting/d696282b-afb9-4aa3-bc1f-3f2c31be5678>

Retail SuperStore

Select a Period:

1 ene 2015 - 30 jun 2015

KPI 1 - Total Sales

\$6,82 M

KPI 2 - Total Profit

\$491,39 mil

KPI 3 - Total Quantity Sold

83,3 mil

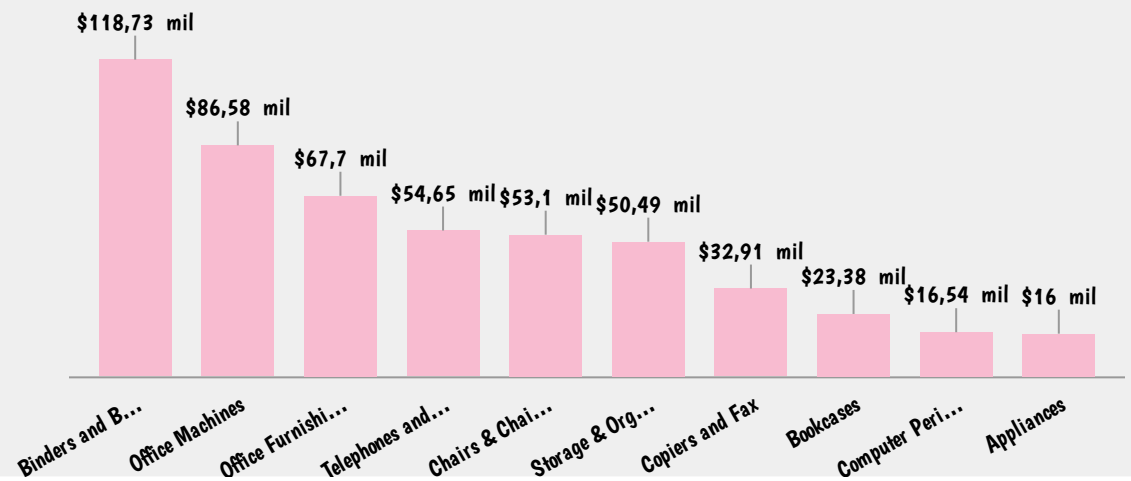
KPI 4 - Average Discount

4,92%

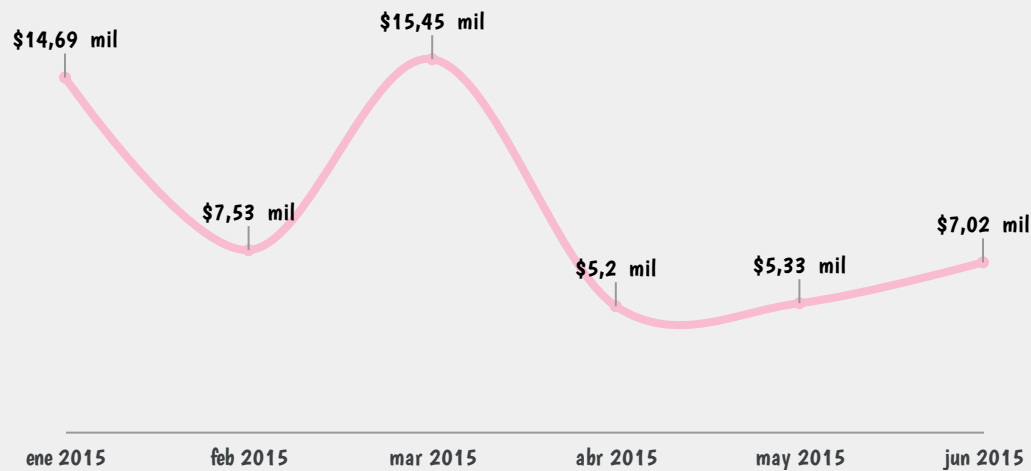
Sales by Product Category



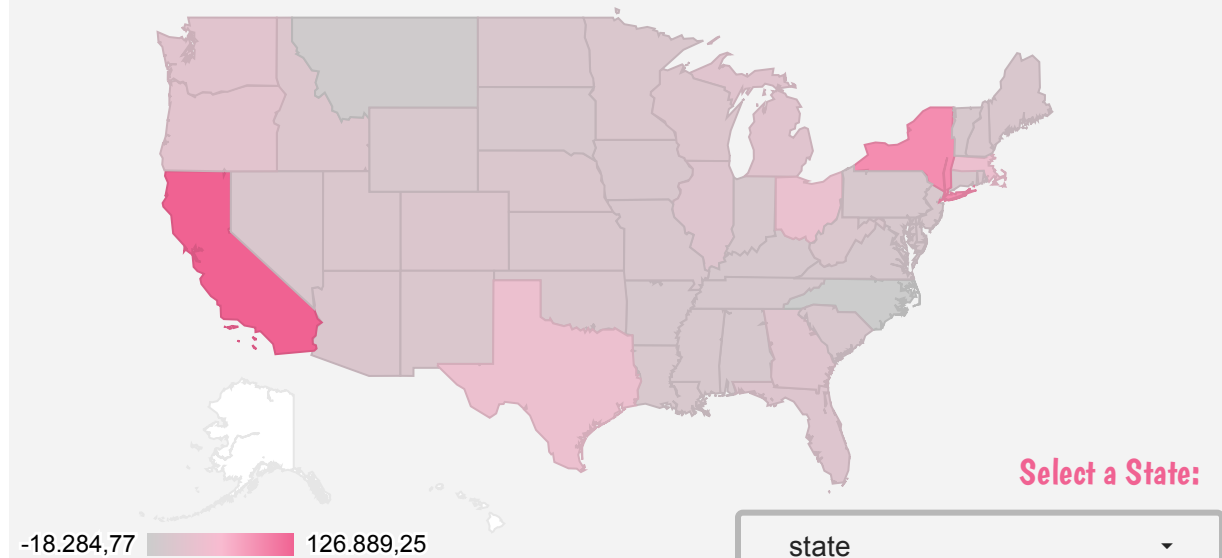
Profit by Sub-Category



Monthly Sales Trend



Profit by State



Appendix B – SQL Scripts for Data Warehouse Creation

This appendix contains the core SQL scripts used to implement the dimensional data warehouse in Google BigQuery.

The scripts demonstrate how raw transactional data from the staging layer is transformed using an ELT approach into structured dimension tables (Date, Product, Customer, and Location) and the central fact table (Fact_Sales).

These queries include the generation of surrogate keys, data type standardisation, deduplication logic, and the establishment of relationships between fact and dimension tables.

The purpose of this appendix is to provide technical transparency, reproducibility of the data model, and evidence of the practical implementation of the proposed architecture. It also serves as a reference for future maintenance, scalability, and automation of the data pipeline.

Google BigQuery Project (Data Warehouse & SQL Processing)

This link provides access to the Google BigQuery environment where the staging tables, dimension tables, fact tables, and transformation queries were implemented and executed as part of the ELT pipeline: <https://console.cloud.google.com/bigquery?project=superstore-483622>

Indicative SQL for SCHEMA Creation

```
CREATE SCHEMA IF NOT EXISTS dw;
```

Indicative SQL for Dimension Table Creation

 Date Dimension

```
CREATE TABLE dw.dim_date AS
SELECT DISTINCT
  CAST(
    FORMAT_DATE('%Y%m%d', DATE(`Order Date`))
    AS INT64
  ) AS date_key,
  DATE(`Order Date`) AS full_date,
  EXTRACT(DAY FROM DATE(`Order Date`)) AS day,
  EXTRACT(MONTH FROM DATE(`Order Date`)) AS month,
  EXTRACT(QUARTER FROM DATE(`Order Date`)) AS quarter,
  EXTRACT(YEAR FROM DATE(`Order Date`)) AS year
FROM staging.superstore_raw
WHERE `Order Date` IS NOT NULL;
```



Product Dimension

```
CREATE TABLE dw.dim_product AS
SELECT
  ROW_NUMBER() OVER () AS product_key,
  `Product Category` AS category,
  `Product Sub-Category` AS sub_category,
  `Product Name` AS product_name

FROM (
  SELECT DISTINCT
    `Product Category`,
    `Product Sub-Category`,
    `Product Name`
  FROM staging.superstore_raw);
```

Customer Dimension

```
CREATE TABLE dw.dim_customer AS
SELECT
  ROW_NUMBER() OVER () AS customer_key,
  `Customer ID` AS customer_id,
  `Customer Name` AS customer_name,
  `Customer Segment` AS customer_segment

FROM (
  SELECT DISTINCT
    `Customer ID`,
    `Customer Name`,
    `Customer Segment`
  FROM staging.superstore_raw);
```

Location Dimension

```
CREATE TABLE dw.dim_location AS
SELECT
  ROW_NUMBER() OVER () AS location_key,
  Country AS country,
  Region AS region,
  `State or Province` AS state,
  City AS city,
  `Postal Code` AS postal_code

FROM (
  SELECT DISTINCT
    Country,
    Region,
    `State or Province`,
    City,
    `Postal Code`
  FROM staging.superstore_raw);
```

Indicative SQL for Fact Table Creation

```
CREATE TABLE dw.fact_sales AS
SELECT
  ROW_NUMBER() OVER () AS sales_id,

  -- Measures
  CAST(`Sales` AS FLOAT64) AS sales_amount,
  CAST(`Profit` AS FLOAT64) AS profit,
  CAST(`Quantity ordered new` AS INT64) AS quantity,
  CAST(`Discount` AS FLOAT64) AS discount,
  CAST(`Shipping Cost` AS FLOAT64) AS shipping_cost,

  -- Foreign Keys
  d.date_key,
  p.product_key,
  c.customer_key,
  l.location_key

FROM staging.superstore_raw r

JOIN dw.dim_date d
  ON r.`Order Date` = d.full_date

JOIN dw.dim_product p
  ON r.`Product Category` = p.category
 AND r.`Product Sub-Category` = p.sub_category
 AND r.`Product Name` = p.product_name

JOIN dw.dim_customer c
  ON r.`Customer ID` = c.customer_id

JOIN dw.dim_location l
  ON r.Country = l.country
 AND r.Region = l.region
 AND r.`State or Province` = l.state
 AND r.City = l.city;
```



References

Han, J., Kamber, M., & Pei, J. (2012). *Data mining: Concepts and techniques* (3rd ed.). Morgan Kaufmann.

Inmon, W. H. (2005). *Building the data warehouse* (4th ed.). Wiley.

ISO/IEC. (2018). *ISO/IEC 25012: Data quality model*. International Organization for Standardization.

Kimball, R., & Ross, M. (2013). *The data warehouse toolkit: The definitive guide to dimensional modeling* (3rd ed.). Wiley.

Marr, B. (2016). *Big data in practice: How 45 successful companies used big data analytics to deliver extraordinary results*. Wiley.

Google Cloud. (2024). *BigQuery ML overview*.
<https://console.cloud.google.com/bigquery?project=superstore-483622>

Google Cloud. (2024). *Looker Studio documentation*.
<https://lookerstudio.google.com/reporting/d696282b-afb9-4aa3-bc1f-3f2c31be5678>

Vivek468. (2023). *Superstore dataset (final)* [Dataset]. Kaggle.
<https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>

Oracle. (2020). *Data warehousing concepts* (White paper). Oracle Corporation.